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The design cycle, end to end

Lecture 1 · CADE30008 Flight Dynamics & Advanced Control

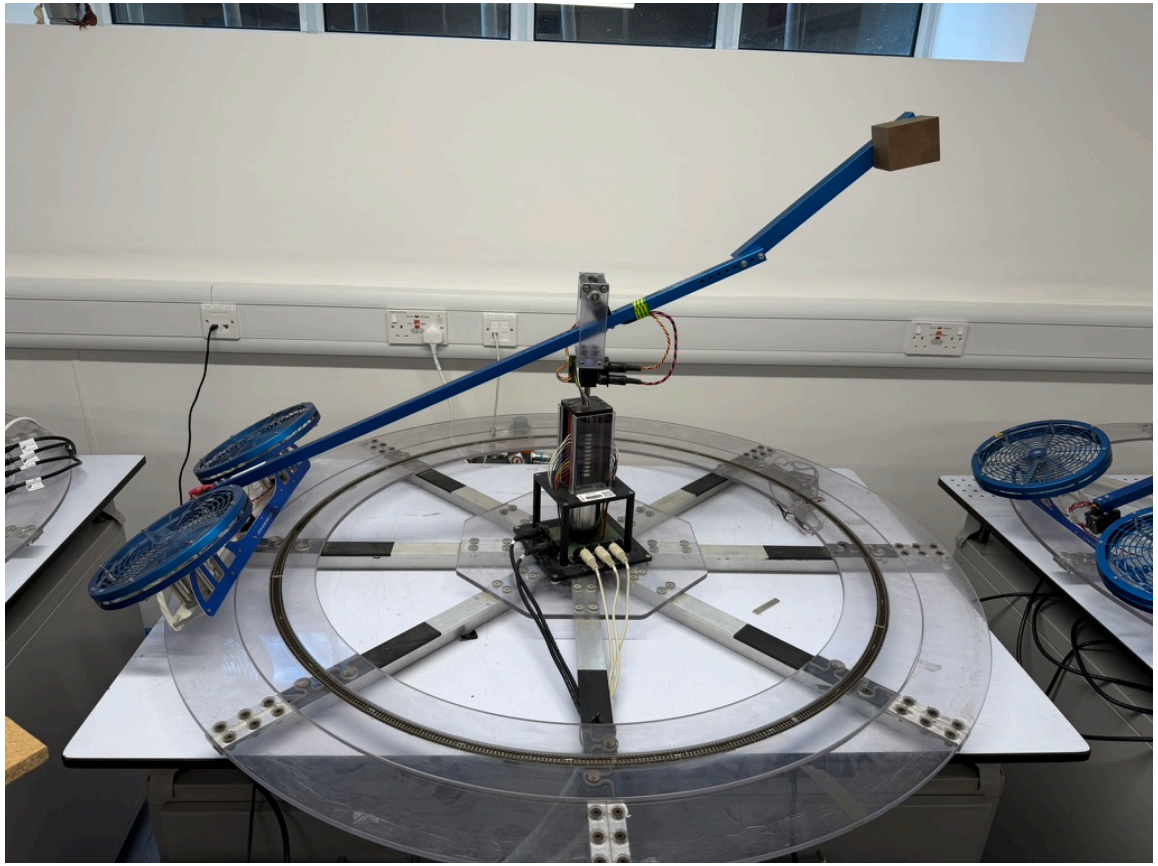
Dr. Steve Bullock

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The claim

Made now, tested in half an hour

This machine cannot be flown by hand



One of you is going to try.

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Why control exists

And the design cycle our methods are part of

The mathematics this unit starts from

- Laplace transforms, transfer functions, poles and zeros
- Standard second-order form: what ζ and ω_n do to a response
- Bode plots from first- and second-order factors
- PID, in its parallel form

If any of that feels distant: the prerequisite diagnostic. Eleven questions, worked feedback on every option, no marks.

Control is a cycle

1. **Understand the plant.** How it performs, ideally not what the datasheet says
2. **State the requirement.** What 'good' looks like, in numbers, before designing
3. **Design.** Structure, gains and other parameters, a prediction
4. **Validate in simulation.** Does it meet the requirement?
5. **Test on hardware.** Does it *really*?
6. **Go round again**, because 5 disagreed with 4

> **Today you do all six. Quickly. Once.** To put the rest of the course in context: everything after today is one of these steps, done properly.

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Feedback

What it gives, and what it costs

Feedback is a trade, not a fix

You get, for free:

- disturbance rejection
- insensitivity to the plant
- the ability to stabilise something unstable

You pay, unavoidably:

- sensor noise reaches the output
- the loop can go unstable, which the open plant could not
- **more gain brings more of both**

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The machine

Three axes, two motors

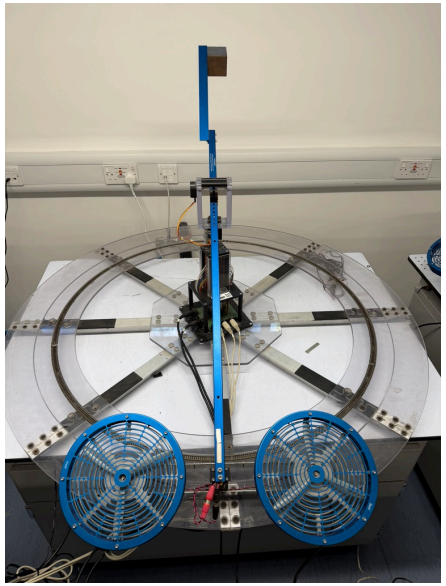
It is a tandem-rotor helicopter





Quanser model it on the **Boeing HC-1B Chinook**. Or, if helicopters are not your thing: **half a quadcopter**.

Elevation, pitch, travel



Elevation ε	height of the beam	zero when level
Pitch ρ	tilt of the motor pair	front motor higher
Travel λ	swing about the column	counter-clockwise from above

Before anything is switched on



The circled rocker switch is the emergency stop. No separate button, no software interlock. Know where it is before the motors spin.

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The attempt

Two joysticks, three axes

YOU DO

Fly it

Open loop. Two joysticks. All three axes.

Take off and fly for 10 seconds. It ends when you hit an axis limit.

Everyone else: count.

Why a person loses to this

- The oscillation **does not die away** on its own
- Every correction arrives **slightly too late**, and makes it bigger
- Fixing elevation with pitch **sends it round the track**
- You are controlling **three coupled things with two levers**

None of this is about reflexes.

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One axis

And a model of it

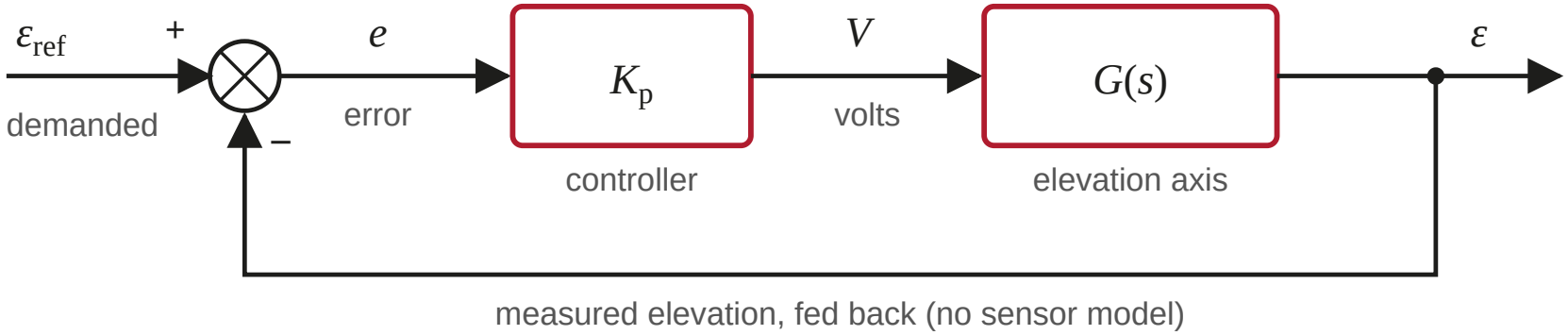
Elevation, measured

$$G(s) = \frac{\varepsilon(s)}{V(s)} = \frac{K\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}s + \omega_{\mathrm{n}}^2}$$

> K ≈ **3.4** deg/V · ω_n ≈ **1.0** rad/s · ζ ≈ **0.06**

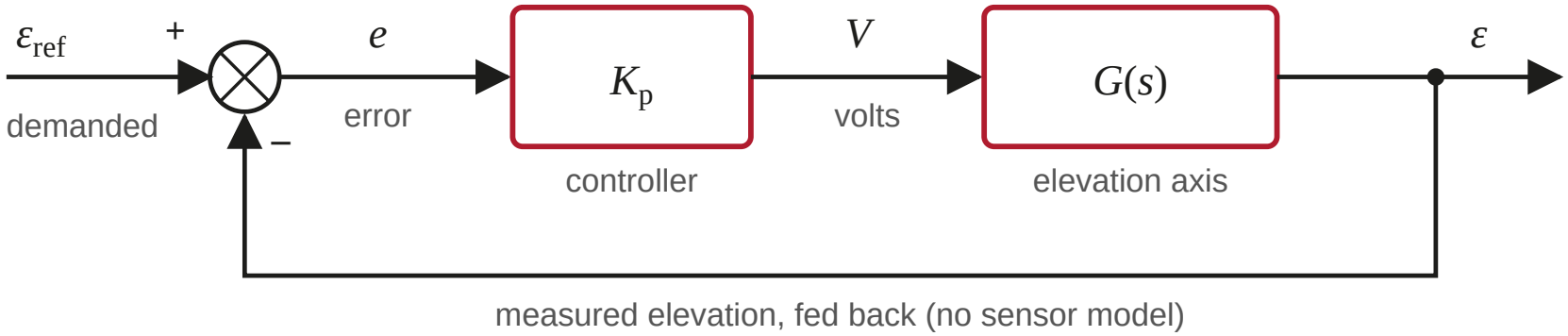
- **Stable.** Disturb it and it does come back
- ζ = 0.06 is **almost no damping**: six-second period, about a minute to settle
- Nudge it and walk away, and it is still moving when you return

So close a loop around it



Measure the elevation, compare it with the one you asked for, and drive the motors depending on the difference.

Reminder: closed-loop TF & characteristic polynomial



$$e = \varepsilon_{\text{ref}} - \varepsilon, \qquad V = K_p e, \qquad \varepsilon = G(s)\, V$$

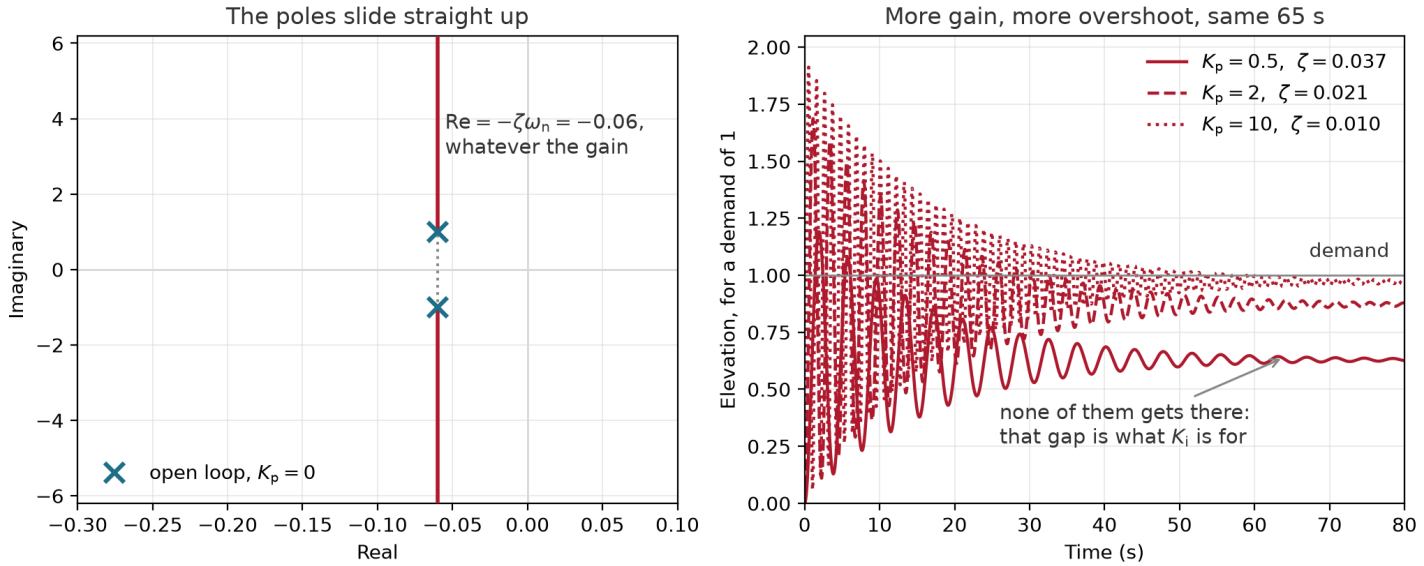
$$\varepsilon\,(1 + K_p G) = K_p G\, \varepsilon_{\text{ref}} \qquad \Longrightarrow \qquad \frac{\varepsilon}{\varepsilon_{\text{ref}}} = \frac{K_p G}{1 + K_p G}$$

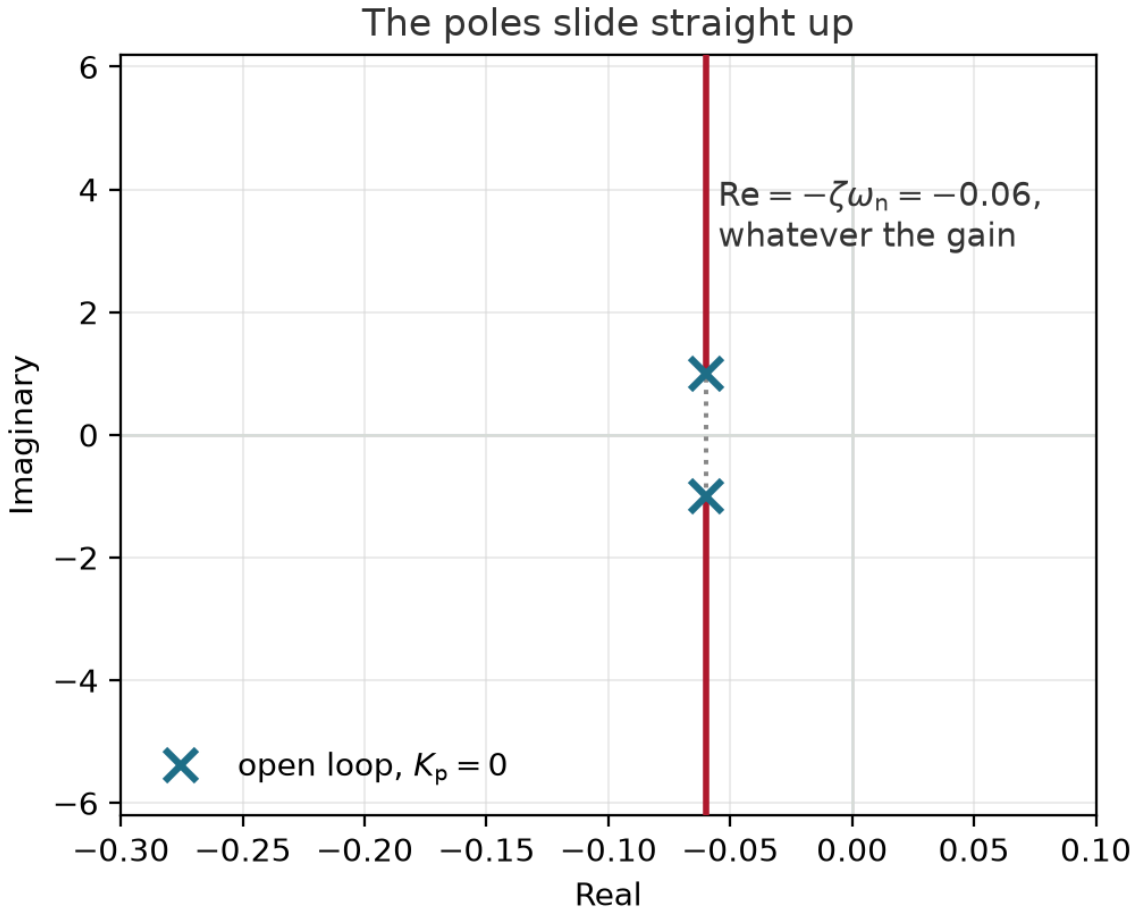
$$G = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \implies \text{denominator} = s^2 + 2\zeta\omega_n s + \omega_n^2\,(1 + K K_p)$$

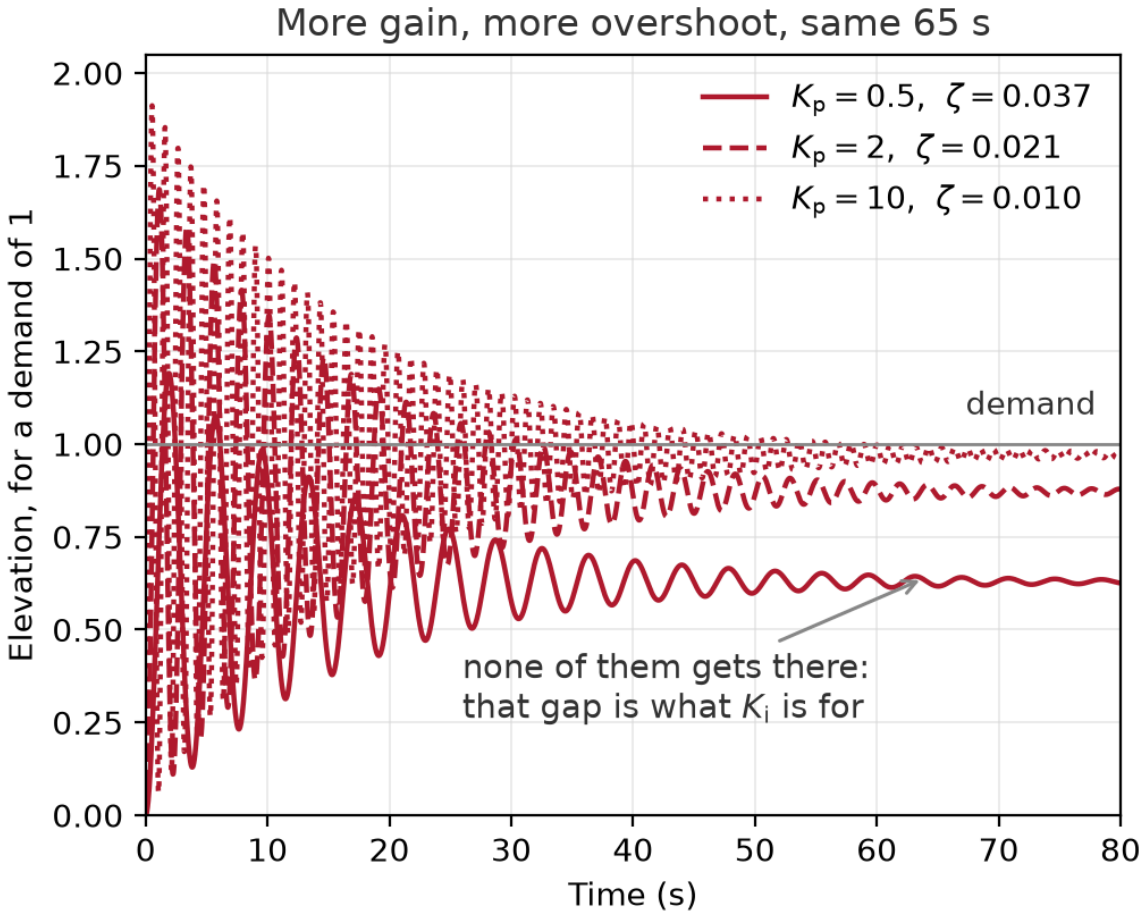
"Turn the gain up" does not work here

$$s^2 + 2\zeta\omega_n s + \omega_n^2(1 + KK_p)$$

K_p is not in the coefficient of s.




The poles slide straight up
Re = −ζω_n = −0.06,
whatever the gain
open loop, K_p = 0
Imaginary
Real
6
4
2
0
−2
−4
−6
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10


More gain, more overshoot, same 65 s
K_p = 0.5, ζ = 0.037
K_p = 2, ζ = 0.021
K_p = 10, ζ = 0.010
demand
none of them gets there:
that gap is what K_i is for
Elevation, for a demand of 1
Time (s)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
10
20
30
40
50
60
70
80

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What counts as good?

Agreeing it before we design

A validatable requirement is a number with a test

Quantity	Means	Typical
Overshoot M_p	how far past it goes	"no more than 20%"
Settling time t_s	until it stays in a band	"within 2% inside 8 s"
Steady-state error	what is left at the end	"under 1 degree"

A pilot's judgement gets you a long way. It cannot be met, missed, or worked with analytically.

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System identification

Getting the model from the machine

YOU DO

Three numbers, three measurements

- **Period** of the oscillation → damped frequency, $\omega_d = 2\pi/T_d$
- **Ratio of successive peaks** → damping ratio, by log decrement
- **Steady value** → the gain

> Do it **by hand first**. `tfest` fits any order in one line, and a fit you can understand and critique is more useful to you.

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Design, and submit

Your gains, on the real machine

YOU DO

Tune it, then send it

- Tune in simulation until it meets **the requirement we agreed**
- `s2_tune` **checks your gains** and gives you a pre-filled link to submit them to me
- You press Submit, so you see what goes out **under your name**

> Gains outside the envelope are **refused, not adjusted**. You will be told which limit you missed.

We'll fly three rounds

1. **The extremes.** Most aggressive, most sluggish, most integral
2. **The cohort average.** Often worse than most of its parts
3. **The best few**

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Simulation, meet hardware

Where they disagree

It met the spec. It still might not fly

- **The model is a simplification.** One axis, linearised, fitted over seconds
- **The actuators saturate.** Ask a 24 V amplifier for 30 V and it will not give it to you
- **There is noise.** The encoder quantises, and D amplifies exactly that
- **The plant moves.** Trim, friction, and the rig at 14:00 is not the rig at 13:00

Each of these gets a week of its own. The answer is to design for it, not to model harder first.

Today, in five lines

- Control design is a **loop**, and every week is one step of it
- Feedback is a **tradeoff**: disturbance rejection against noise and instability
- A **requirement** is a number with a test, agreed *before* designing
- A model you **measured** beats a model you assumed, and knowing which you have matters
- **Simulation and hardware often disagree**, and we can design for that

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There is no right answer until you say what you want

See you next lecture